



Use Your Smartphone As A Smart Remote For TV And Set-top Box

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Smartphones have become an integral part of our lives but when it comes to watching our favourite TV shows, we have to look for the TV and set-top box remotes. Why not build the functionality of a TV and a set-top box remote in your smartphone? This project explores the possibility.

How a remote works

The functional diagram of a remote controller is shown in Fig. 1, which has following blocks.

The keypad. The remote has several keys which, when pressed, instruct the TV/set-top box to perform a specified function. On pressing a key, the key-scan code for that key is passed on to an encoder.

The encoder. You would have noticed that the remote of one brand or product does not work for another brand. For example, you cannot use the remote of a Sony TV to operate an LG TV. This happens as each brand of remote assigns a unique code to its keys.

The remote controller is a serial device that transmits the code bits serially, one

bit after the other. The sequence in which these bits are transmitted is also important. For example, while transmitting code 100100, the bits can either be sent out as 1-0-0-1-0-0 or in the reverse order as 0-0-1-0-0-1. This gets a bit complicated when additional bits like start bit and end bit, error correction, etc have to be added. All such rules together are called a protocol.

The encoder translates the key-scan code to the manufacturer-specified corresponding code and packages it as per the prescribed protocol.

The carrier generator and modulator. A remote controller can use radio waves, Bluetooth, Wi-Fi, or infrared (IR) waves to communicate with the TV set. Most remotes use IR waves.

Infrared is part of the electromagnetic wave spectrum and very similar to light, though with a higher wavelength. This similarity causes problems as infrared waves are present everywhere, be it sunlight or a warm body.

The solution lies in using a carrier wave to carry the message. IR remotes use a carrier wave in the 30-40MHz range, most common being the 38MHz frequency.

The code being sent is made to piggy-back on this carrier wave by way of pulse modulation. The TV receives these waves and recovers the remote code through a demodulator.

The process is similar to radio signals where the audio signals piggy-back on a radio frequency carrier wave and after the radio receiver tunes in on to them, demodulates it, restoring the audio signals.

The IR emitter. The IR emitter is an infrared LED that switches on and off at 38MHz in sequence with the carrier wave, thereby generating a pulsating infrared wave. This signal is received by the TV, which has an infrared receiver. The signal is demodulated by removing the 38MHz carrier and the code transmitted by the remote is recovered. Action is now taken based on the code transmitted. The IR signal transmission is shown in Fig. 2.

Project outline

In the current project, we shall be making a Wi-Fi enabled remote controller which would replace the TV and set-top box remote and add greater functionality to it. We shall not be making any physical changes to the mobile phone or TV set. The functional diagram of our Wi-Fi remote is shown in Fig. 3.

The project involves building an Android app and decoding the TV/set-top box remote.

Android app development

The Android app is built using MIT App Inventor. Follow the steps below to download and install the app on your mobile.

1. Download the file *TVRemote_Ver1_4.apk* from EFY site (source.efymag.com) to your mobile phone.

2. Go to Security settings on your phone and enable 'Allow installation of apk files from unknown sources.' Since this program is built by us only, there is no risk of loading it. Click on the *TVRemote_Ver1_4.apk* file in your

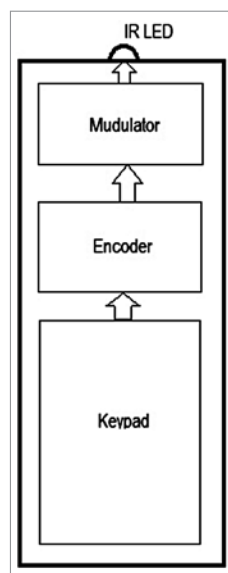


Fig. 1: Functional diagram of a remote

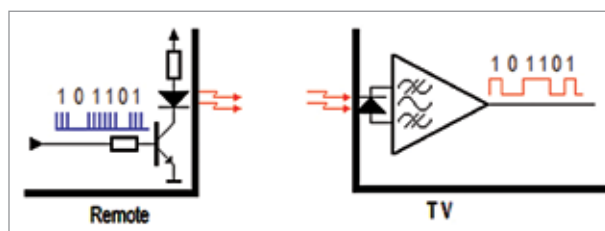


Fig. 2: IR signal transmission